

CLASS TEST

1) An image appears too bright. Explain the most suitable grey level transformation

- ◆ Grey level Transformation: Negative or power-law
- ◆ Concept: Reduces pixel intensity values
- ◆ Application: Apply gamma correction to compress high-intensity pixels
- ◆ Working: Bright pixels become darker while preserving image details
- ◆ Expected outcome: Better visibility with reduced overexposure and improved contrast

2) A dark image lacks details. Describe a suitable grey level transformation

- ◆ Technique: log Transformation
- ◆ Concept: Expands low intensity pixel values
- ◆ Working: Applies the formula

$$S = C \log(1+r)$$

- ◆ Application: Brightens dark region
- ◆ Outcome: Hidden details become visible without over brightening the image

3) An image has poor contrast. Explain a histogram based enhancement technique

- ◆ Technique: Histogram Equalization
- ◆ Concept: Redistributes intensity values

- ◆ Working: Spreads pixel values across the full intensity range
- ◆ Application: Enhances Contrast
- ◆ Outcome: Object becomes clear and image quality improves

4) A Noisy image contains random intensity variations
Explain the best smoothing filter

- ◆ Technique: Gaussian filter
- ◆ Concept: Removes Gaussian Noise
- ◆ Working: uses a weighted average of neighboring pixels
- ◆ Application: smooths the image while preserving overall
- ◆ Outcome: Noise is reduced and the image becomes clear

5) An image appears blurred after smoothing
Explain a sharpening method;

- ◆ Technique: unsharp Masking
- ◆ Concept: Enhances edges
- ◆ Working: Subtracts a blurred image from the original and adds the difference back
- ◆ Application: Restores lost details
- ◆ Outcome: Sharper edges and improved image clarity

6) An image contains high-frequency noise
Explain the appropriate frequency domain filter

- Technique: Low-pass filter
- Concept: Removes high-frequency components
- Working: Blocks high-frequency noise while allowing low frequencies to pass
- Application: Smooths noisy images
- Outcome: Reduced noise with preserved image structure

7) An object is clearly brighter than the background
Explain the Segmentation Technique

- ◆ Technique: Thresholding
- ◆ Concept: Separates object and background using intensity
- ◆ Working: Pixels above the threshold become the Object; others become the background
- ◆ Application: Simple image Segmentation
- ◆ Outcome: Accurate separation of the object from the background

8) An image requires identification of Object edges
Explain a suitable edge detection Method

- ◆ Technique: Canny Edge Detection

- ◆ Concept : Detects strong and continuous edges
- ◆ working : uses Gaussian smoothing, gradient calculation, non-maximum suppression and hysteresis thresholding
- ◆ Application: Detects object boundaries accurately
- ◆ Outcome : Thin, continuous, and noise-free edges

9) An image contains salt and pepper noise. Explain the appropriate spatial filter

- ◆ Technique : Median filter
- ◆ Concept : Removes impulse noise
- ◆ working : Replaces each pixel with the median value of its neighborhood
- ◆ Application: Eliminates black and white noise
- ◆ Outcomes : Noise is removed while preserving image edges

10) A gray scale image improve poor separation between object and background. Explain a thresholding approach

- ◆ Technique : Otsu's Thresholding.
- ◆ Concept : Automatically select the best threshold
- ◆ working : minimize intra-class variance between foreground and background
- ◆ Application: Segments images without manual threshold selection
- ◆ Outcome : Accurate object - background separation and improved segmentation